

timeline check-in

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Set up

Packages

```
# general use and cleaning
library(tidyverse)
library(here)
library(janitor)
library(snakecase)

# wrapping axis labels
library(scales)

# reading in .xlsx files
library(readxl)

# arranging plots
library(patchwork)
```

Data

```
# taxonomic information
taxon_list <- read_csv(here("data", "taxon_list.csv"))

# aquatic invertebrates
aq_ins <- read_xlsx(here("data", "Aquatic Sampling Data-2026-03-10.xlsx"),
                   sheet = "Aquatic Insects")

# site information
```

```
sites <- read_xlsx(here("data", "Aquatic Sampling Data-2026-03-10.xlsx"),
  sheet = "sites")

# water quality information
water_quality <- read_xlsx(here("data", "Aquatic Sampling
  ↪ Data-2026-03-10.xlsx"),
  sheet = "Water Quality",
  na = c("", "n/a", "N/A", "over", "173+"))
```

Cleaning and Wrangling

Cleaning

sites object

```
# creating new object from sites
ncos_sites <- sites |>
  # filter to only include NCOS sites sampled four times a year
  filter(sector == "NCOS" & sampling_frequency == "Quarterly")
```

taxon_list object

```
# creating new clean object from taxon_list
taxon_list_clean <- taxon_list |>
  # converting taxon name to snake case (no spaces or capital letters,
  # only underscores)
  mutate(verbatim_name = to_any_case(verbatim_name, case = "snake")) |>
  select(verbatim_name, scientificName)
```

water_quality object

```
water_quality_clean <- water_quality |>
  clean_names() |>
  semi_join(ncos_sites, by = "site") |>
  unite("date_site", date, site, remove = TRUE) |>
  group_by(date_site) |>
  summarize(
    med_ph = median(p_h, na.rm = TRUE),
    med_sal = median(salinity_ppt, na.rm = TRUE),
```

```

        med_do      = median(dissolved_oxygen_mg_l, na.rm = TRUE),
        med_temp    = median(temperature_c, na.rm = TRUE)) |>  # <-- ADD
        ↪ THIS
ungroup() |>
filter(date_site != "2022-08-12_NVBR")

```

aq_ins object

```

# creating new clean object from aquatic inverts
aq_ins_clean <- aq_ins |>
# clean column names
clean_names() |>
# filtering join: only include sites occurring in ncos_sites
semi_join(ncos_sites, by = c("site")) |>
# renaming columns
rename(date = date_on_vial) |>
# select columns of interest
select(date, sample_type, site,
        ostracod,
        copepod,
        cladocera) |>
# filter to only include "filtered beaker" samples
filter(sample_type %in% c("FB 250", "FB250")) |>
# convert the data frame to long format
pivot_longer(cols = ostracod:cladocera,
              names_to = "taxon",
              values_to = "abundance") |>
# group by date, site, taxon
group_by(date, site, taxon) |>
# calculate average abundance per liter (sum abundance divided by 7.5
  ↪ liters)
summarize(ave_lit = sum(abundance, na.rm = TRUE)/7.5) |>
# ungroup data frame
ungroup() |>
# create a new column called `date_site` to join with water quality
unite("date_site", date, site, remove = FALSE) |>
# join with water quality
left_join(water_quality_clean, by = c("date_site")) |>
# join with taxon list
left_join(taxon_list_clean, by = c("taxon" = "verbatim_name")) |>
# add full names of sites

```

```

mutate(site_full = case_when(
  site == "NVBR" ~ "North of Venoco Bridge",
  site == "NEC" ~ "South of Venoco Bridge",
  site == "NMC" ~ "Main Channel",
  site == "NPB" ~ "South of Phelps Bridge",
  site == "NPB1" ~ "North of Phelps Bridge",
  site == "NPB2" ~ "Phelps Road",
  site == "NWP" ~ "West Pond",
  site == "NDC" ~ "Devereux Creek"
)) |>
# setting factor levels for site
mutate(site_full = fct_reorder(site_full, med_sal, .fun = "median", .na_rm
  ↪   = TRUE),
  site_full = fct_rev(site_full)) |>

# ADD: log+1 transform of average abundance per liter
mutate(log_ave_lit = log(ave_lit + 1)) |> # <-- Fig
  ↪   1, 2, 3, 4

# ADD: season derived from date
mutate(season = case_when(
  month(date) %in% c(3, 4, 5) ~ "Spring",
  month(date) %in% c(6, 7, 8) ~ "Summer",
  month(date) %in% c(9, 10, 11) ~ "Fall",
  month(date) %in% c(12, 1, 2) ~ "Winter"
),
season = factor(season, levels = c("Spring", "Summer", "Fall", "Winter")))
  ↪   |># <-- Fig 4
mutate(taxon = factor(taxon, levels = c("copepod", "ostracod",
  ↪   "cladocera")))

```

Setting plot aesthetics

```

# colors for taxa
taxon_cols <- c(
  "ostracod" = "orange2",
  "copepod" = "tomato3",
  "cladocera" = "turquoise4")

sites_cols <- c(

```

```

"NDC" = "hotpink3",
"NEC" = "cyan3",
"NMC" = "lightpink2",
"NPB" = "slateblue3",
"NPB1" = "steelblue3",
"NPB2" = "green3",
"NVBR" = "maroon3"

)

# setting common theme elements
theme_set(
  theme_classic() +
    theme(text = element_text(size = 12),
          plot.title.position = "plot",
          legend.position = "none"))

```

Visualizations directly relevant to answering questions

Figure 1

```

figure1 <- ggplot(data = aq_ins_clean,
                  mapping = aes(x = med_temp,
                                y = log_ave_lit,
                                color = taxon)) +

  geom_point() +

  facet_wrap(~site_full) +
  labs(title = "Relationship Between Water Temperature and Invertebrate
    ↪ Abundance by Species and Site",
       x = "Average Temperature (°C)",
       y = "log + 1 Transformed Average Abundance/L",
       color = "Taxon") +

  scale_color_manual(values = taxon_cols) +

  theme(legend.position = "right")

```

figure1

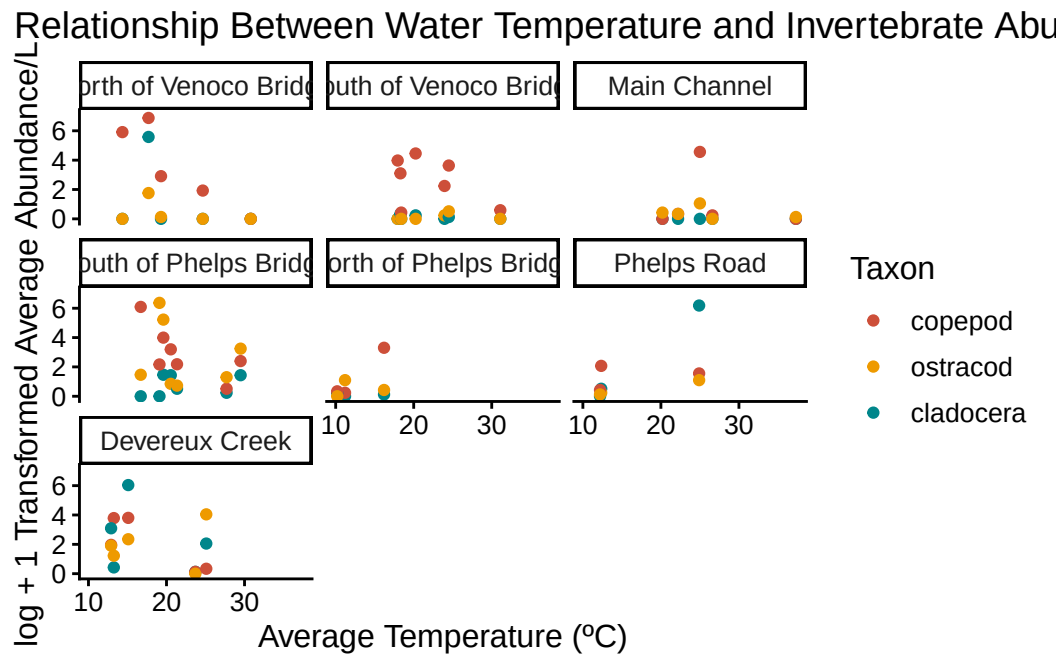


Figure 2

```
figure2 <- ggplot(data = aq_ins_clean,
  aes(x = taxon,
    y = log_ave_lit,
    color = taxon)) +

  geom_jitter(height = 0,
    alpha = 1,
    width = 0.2,
    shape = 21) +
  # second layer: points to represent medians
  stat_summary(geom = "point",
    fun = "mean",
    size = 4,
    color = "black",
    shape = 18) + # 18 = filled diamond
```

```

stat_summary(geom = "errorbar",
             fun.data = mean_se,
             width = 0.1,
             color = "black") +

scale_color_manual(values = taxon_cols) +
labs(x = "Taxon",
     y = "log + 1 transformed average abundance/L",
     title = " Mean and Distribution of Invertebrate Abundance (log +1/L)
for Three Tax")

```

figure2

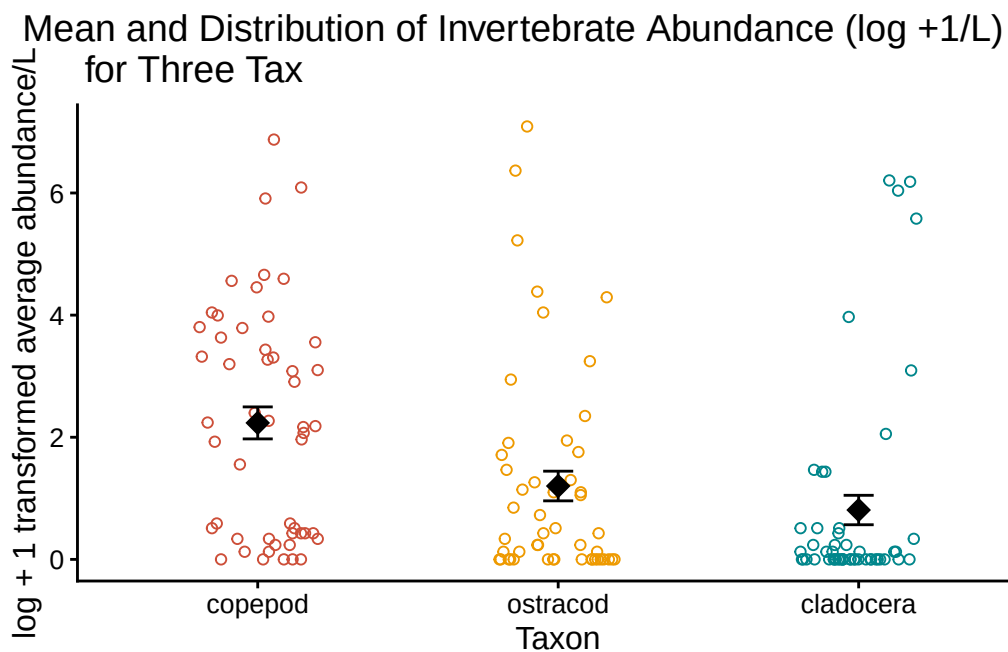


Figure 3

```

# create new object to store figure 4
figure3 <- ggplot(data = aq_ins_clean,
                  mapping = aes(x = med_temp,
                               y = log_ave_lit,
                               color = site)) +

```

```

geom_point() +

facet_wrap(~taxon,
           scales = "free") +

theme(legend.position = "right") +

scale_color_manual(values = sites_cols) +

labs(title = " Site Patterns Between Water Temperature and Log + 1
  Abundance for Three Dominant Taxa",
      x = "Average Temperature (°C)",
      y = "log + 1 Transformed Average Abundance/L",
      color = "Site")

```

figure3

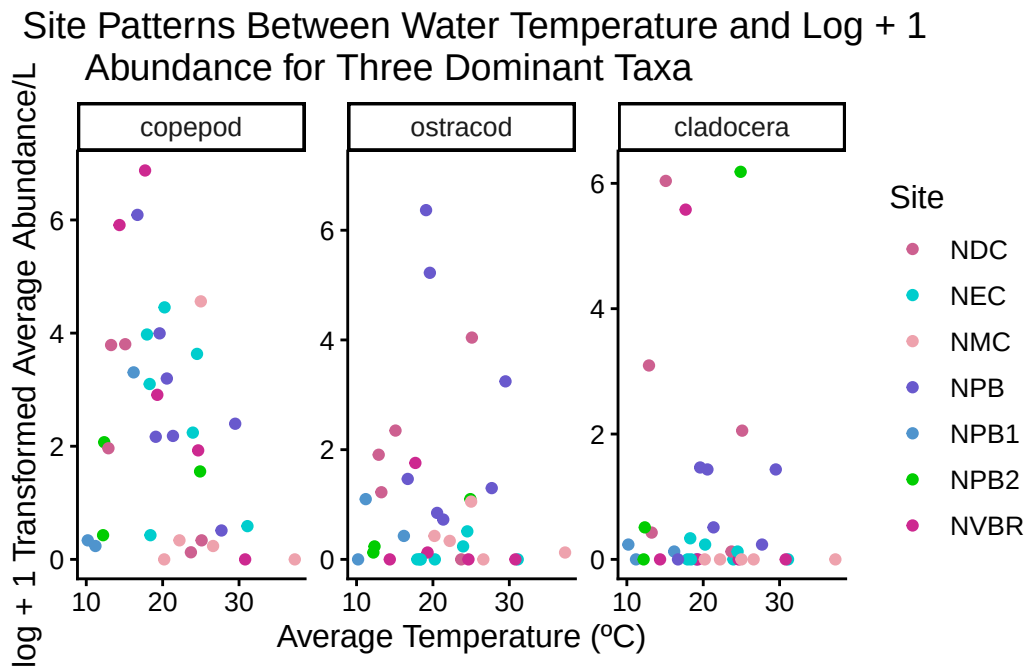


Figure 4

```
# create new object to store figure 4
figure4 <- ggplot(data = aq_ins_clean,
                  mapping = aes(x = med_temp,
                                y = log_ave_lit,
                                color = taxon)) +

  geom_point() +

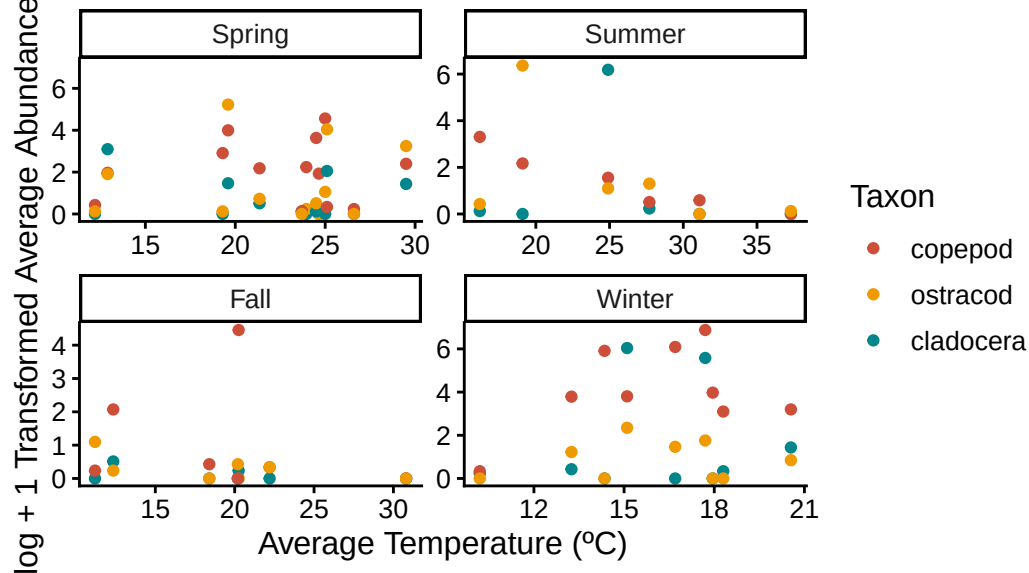
  facet_wrap(~season,
             scales = "free") +
  labs(title = " Seasonal Patterns Between Water Temperature and Log + 1
    Abundance for Three Dominant Taxa",
       x = "Average Temperature (°C)",
       y = "log + 1 Transformed Average Abundance/L",
       color = "Taxon") +

  scale_color_manual(values = taxon_cols) +

  theme(legend.position = "right")

figure4
```

Seasonal Patterns Between Water Temperature and Log + 1 Abundance for Three Dominant Taxa



Other exploratory visualizations

BELOW PASTE IN THE OTHER EXPLORATORY VISUALS

exploring

Plan for elective (if completing)

Timeline

Week 7:

- Research all three organisms (copepod, ostracod, cladoceran)
- Gather reference images
- Review your figures
- Take notes on what each figure shows about the species
- Brainstorm collage/drawing style and decide on materials (digital vs. physical).

Week 8 * timeline check-in*:

- Have a rough sketch/draft of all three 2D collages– one per organism
- Bring your notes on what the figures show about each species. This is our plan and draft milestone.

Week 9:

- Refine and finalize the artwork for 1–2 of the organisms
- Fill in your species notes with complete observations from your figures.

Week 10:

- Complete the third organism's artwork and finish all species notes
- Do a full review, make sure each collage clearly reflects what your figures show about that species.

Finals week:

- Final polish, compile everything together
- Present your completed 2D collage triptych with notes.